Matthew Emuna

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Work and Research Experience

Bridgify.io, Tel-Aviv, Israel *May 2023-August 2023*
Software Engineer Intern
- Delivered 6+ production features on a client-facing travel platform, driving measurable improvements in engagement and performance by building full-stack React and TypeScript (TSX) components within an Agile development workflow.
- Increased international user accessibility by implementing multilingual (i18n) support and redesigning site navigation, improving usability across city and category pages for a global audience.
- Boosted organic search discoverability by restructuring internal linking, replacing JavaScript-rendered links with crawlable HTML hyperlinks, and applying schema.org markup and canonical tags to improve search engine results page (SERP) visibility.
- Reduced page load time by 20% by implementing lazy loading for large images, configuring browser caching policies, and optimizing JavaScript and TypeScript bundles via compression

Skidmore College, Saratoga Springs, New York *May 2025-May2026*
Research Assistant, CNNs, ML Theory
- Surveyed 9 data augmentation strategies (CutMix, YOCO, Elastic Deformation, geometric transforms) for brain MRI tumor classification on BraTS 2018, training a multi-backbone Convolutional Neural Network (CNN).
- Developed KAug, a novel augmentation leveraging K-means clustering for unsupervised tumor-region segmentation and targeted intensity perturbation — achieving 90.1% test accuracy, outperforming 7 of 9 baseline methods.

Projects

PL xG Predictor [https://premier-league-xg-predictor.vercel.app/] *June 2026-Present*
Python, FastAPI, scikit-learn, PostgreSQL, React, Vite, Tailwind, Anthropic API, REST
- Built an end-to-end football analytics platform predicting expected goals (xG) for any matchup, serving real-time model inference and AI-generated analysis through a FastAPI REST backend and a React/Vite frontend.
- Engineered a 5-season, 27-club data pipeline in Python/PostgreSQL with leakage-safe rolling features, assembling a match-level training set enriched with ClubElo ratings and head-to-head history.
- Trained dual scikit-learn LinearRegression models to predict home and away xG, achieving a 0.6 mean absolute error.
- Integrated an agentic LLM analyst using the Anthropic API (Claude) with a custom tool-use loop, enabling the model to call the xG predictor mid-conversation and return structured predictions rendered as interactive frontend cards.

Mancala AI Agent *January 2026-April 2026*
Python, Reinforcement Learning, Minimax, Alpha-Beta Pruning, Monte Carlo Tree Search, Neural Networks
- Engineered a Minimax agent with Alpha-Beta pruning and a custom multi-factor evaluation heuristic (store differential, capture potential, extra turn detection), achieving optimal play and eliminating all losing outcomes against deterministic opponents.
- Developed a reinforcement learning agent using SARSA with linear function approximation, trained over 10,000,000 episodes against both self-play and an Alpha-Beta opponent to improve generalization
- Integrated a PyTorch neural network with SARSA to learn non-linear board evaluation, trained for 1,000,000+ episodes against an Alpha-Beta pruning opponent achieving competitive play

CNN-Based Insect Audio Identification *May 2025-Present*
Python, TensorFlow, Keras, Librosa, SwiftUI, Figma, Data Augmentation
- Collected and processed 1000+ insect audio samples, generating spectrogram images with Librosa for CNN training.
- Achieved 94+% classification accuracy for katydid, field cricket, and snowy tree cricket sounds using a multi-layer CNN with batch normalization, dropout, and dense layers.
- Shipped a functional iOS MVP integrating real-time insect sound recognition, reducing identification time from manual lookup to under 2 seconds, built end-to-end using SwiftUI and Figma prototyping.

Image Inpainting Algorithm *January 2023-May 2023*
Python, OpenCV, Statistical Analysis
- Removed unwanted objects from multi-frame image stacks by applying statistical reconstruction techniques (mean, median, mode, Jenks' natural breaks), improving reconstruction consistency across frames.
- Implemented exemplar-based inpainting to replace missing regions with structurally similar patches, improving continuity in complex textures.

Education

Skidmore College, Saratoga Springs, New York *May 2026*
B.A. Computer Science; Minor: Mathematics | GPA: 3.575

Skills

- **Domains:** Data Structures, Algorithms, Computer Vision, Machine Learning, Neural Networks, Data Augmentation, LLMs, Databases, Applied AI, Backend Development, API Design, System Design, Web Development, Frontend Design
- **Languages & Libraries:** Python, Java, C, Clojure, Swift, JavaScript, TypeScript, SQL, HTML/CSS, TensorFlow, PyTorch, Keras, OpenCV, scikit-learn, Hugging Face Transformers, LangChain, NumPy, Pandas, Matplotlib, Librosa, $\LaTeX$
- **Frameworks & Tools:** React, React Native, Node.js, Next.js, FastAPI, REST, Docker, Kubernetes, Git, Jenkins, CI/CD, PostgreSQL, Snowflake, Xcode, Figma, Claude Code, Cursor, GitHub Copilot, Codex, Twilio, Anthropic API, OpenAI API